

Teaching a Computer to Find Cancer: What Three Training Runs Taught Us

Project: Optimising PSMA PET Segmentation using the Mamba Architecture

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Progress report - 11 August 2026 - covers training runs 1, 2 and 3

What this report is

This is a plain-English progress report on an honours research project. The project is teaching an artificial intelligence model to automatically find and outline prostate cancer tumours in whole-body medical scans. This report covers the last few weeks of work: three complete training runs, the experiments we ran between them, what went wrong, what we fixed, and how the results improved each time. No background in computing or medicine is assumed - every technical idea is explained as it appears.

1. The project in one page

Prostate cancer can spread through the body. A special scan called a PSMA PET scan makes cancer deposits light up like beacons. Right now, a specialist has to look through the whole 3D scan and outline every tumour by hand. That is slow, and two experts will often draw slightly different outlines on the same scan.

Our goal is a computer model that draws those outlines automatically - faster than a person and the same way every time. The hard part is that tumours can be tiny specks inside an enormous image, and parts of the body glow naturally on these scans even when they are perfectly healthy. The model has to catch the specks without constantly crying wolf.

We are building the model with a new architecture called Mamba, and comparing it honestly against the current best published tool for this job, which is called nnU-Net. Throughout this report, 'the reference' means that published tool's results on the same kind of data.

2. What a 'training run' actually is

The model learns by practice. We show it a piece of a scan, it guesses where the tumours are, we tell it how wrong it was, and it adjusts itself slightly. One of those cycles is called an iteration. A full training run in this project is 250,000 iterations - a quarter of a million practice attempts - which takes between 12 and 40 hours of continuous computing on a gaming graphics card. While it trains, we regularly pause and test it on 57 scans it has never seen. That held-out test is how we know whether it is genuinely learning or just memorising.

3. Run 1 - the rocky start

The first full training run finished, but it was troubled. Learning would go well for a while,

then suddenly collapse, then slowly recover, over and over. When we dug into the numbers we found the cause: every so often - roughly one practice attempt in a hundred - the model received a gigantic correction signal, thousands of times bigger than normal. One single measurement spiked to 4,165 when the normal working range is below one. A jolt like that scrambles everything the model has carefully learned, like slamming the steering wheel mid-corner.

Run 1's final quality score was 0.385 on our overlap measure (explained in section 6 - roughly, how well the model's outlines overlap the true ones, where 1.0 is perfect). Frustratingly, run 1's saved model was accidentally destroyed by a test script before it was backed up - a mistake that shaped how carefully we archive everything now.

4. Run 2 - the stabiliser

The fix for run 1's chaos is called gradient clipping. In plain terms, it is a speed limiter on how hard any single practice attempt is allowed to yank the model. Corrections bigger than a chosen limit get scaled down to the limit; everything else passes through untouched. Only 1.27 percent of run 2's quarter-million steps needed limiting - but those were exactly the steps that had been wrecking run 1. The collapses disappeared completely.

Run 2's overlap score rose to 0.514, and for the first time we ran a proper lesion-by-lesion evaluation - not just 'how much do the outlines overlap' but 'of the individual tumours that exist, how many did you find, and how much of what you flagged was real'. That evaluation told us the model was already finding nearly as many tumours as the reference, but flagging too many things that were not tumours. It also revealed a suspect: our scans were being processed at 4-millimetre resolution, where a typical small lesion is only about 19 pixels in size - a smudge barely three pixels across. You cannot outline what you can barely see.

5. Run 3 - sharper eyes

Run 3 made three deliberate changes at once. First, we doubled the sharpness: the scan is now processed at roughly 2-millimetre resolution, so the same small lesion becomes about 100 pixels instead of 19 - five times more detail on exactly the tumours the model was getting wrong. Second, because sharper pixels shrink the model's field of view, we enlarged its viewing window so it still sees the same amount of the body at once. Third, we rebalanced the penalty system: in run 2, missing a tumour was punished about 2.3 times harder than raising a false alarm, which taught the model to flag anything remotely suspicious; run 3 made both mistakes cost the same.

Run 3 also survived a mishap worth telling. Two hours in, the training process was abruptly killed. The detective work traced it to a memory problem: at the first mid-training test, the system spun up six extra data-loading helpers, each holding entire full-body scans - which at the new, sharper resolution are eight times larger than before. Together they exhausted the computer's memory and the operating system killed the run to protect itself. The fix was one line of code, and because the model's progress is saved

at every test point, we resumed with only 50 iterations lost - about 20 seconds of progress. The remaining 38 hours ran without a single hiccup, and training was the calmest yet: only 0.69 percent of steps needed the speed limiter, half of run 2's rate.

6. How we measure success - the four numbers that matter

Before the results, here is what each measurement actually means. Nothing else in this report makes sense without these.

- Dice score (also called the overlap score): how well the model's outlines overlap the true outlines, from 0 (no overlap) to 1 (perfect). Good for tracking training, but it is dominated by big tumours - a model can score decently while missing many small ones.
- Sensitivity (also called recall): of the tumours that really exist, what fraction did the model find? This is the clinical priority - a missed tumour is worse than a false alarm a doctor can quickly dismiss.
- Precision (also called PPV): of the things the model flagged, what fraction were real tumours? Low precision means the doctor wastes time rejecting suggestions, and stops trusting the tool.
- F1 score: a single number that balances sensitivity and precision. This is the headline number the reference tool is quoted in, so it is our headline too.

7. The results, run by run

The overlap score climbed every run: 0.385 in run 1, then 0.514 in run 2, then 0.545 in run 3. And importantly, run 2 and run 3 were tested using exactly the same protocol, so that last improvement is a clean, fair comparison.

The lesion-level results are where it gets interesting. Comparing run 3 to run 2 on the standard evaluation: the model found 108 more real tumours (1,286 versus 1,178) and missed 95 fewer. Sensitivity - the fraction of real tumours found - jumped from 66.7 percent to 72.3 percent. And before any filtering, run 3's sensitivity reached 73.5 percent, which is actually past the reference tool's published 73.0 percent - the first time this project has beaten the reference on any measure. That is remarkable given our model trains on less than half the data the reference used.

The sharper resolution also fixed a specific bad habit. At 4 millimetres, the model would often blur several neighbouring tumours into one big blob - common in this cancer, where chains of small deposits sit close together. We can measure that habit in the data, and it roughly halved at 2 millimetres. The sharper eyes genuinely separate what the blurry ones fused together.

8. The rubber ruler - a discovery about our own measuring stick

One number appeared to get worse in run 3: precision dipped from 74.0 percent to 70.4

percent on the official protocol. Investigating that dip led to the most important discovery of this update - and the problem turned out to be in our measuring stick, not our model.

Our evaluation protocol throws away any predicted tumour smaller than 10 pixels, on the logic that specks that tiny are usually noise. Here is the catch: a 'pixel' is not a fixed physical size. At run 2's 4-millimetre resolution, 10 pixels is about 663 cubic millimetres of tissue. At run 3's sharper 2-millimetre resolution, 10 pixels is only about 124 cubic millimetres - so the same rule filters five times less aggressively. Run 3 was being judged with a far weaker noise filter than run 2, purely as a side effect of the resolution change. We were measuring the two runs with a rubber ruler.

When we match the filters physically - comparing at roughly the same real-world volume - run 3 beats run 2 on every single measure at once: F1 72.4 versus 70.2 percent, precision 75.6 versus 74.0 percent, and sensitivity 69.5 versus 66.7 percent. In other words, the sharper resolution was an across-the-board win, and the apparent precision drop was an artefact of the measurement. Before the next run, the protocol will be restated in physical units - cubic millimetres - so this can never distort a comparison again.

9. What is still hard - the honest list

First, false alarms on healthy scans. Our test set includes 7 scans with no cancer at all. Run 2 kept 3 of them completely clean; run 3 kept none clean, scattering about 25 false alarms across them. Sharper resolution means eight times more pixels where the model can mistake natural glow for disease - the bladder, kidneys and salivary glands all light up normally on these scans. The reference tool solves this with an anatomy-aware cleanup step that knows where those organs are. We have not built that yet, and this result makes it the clear next priority.

Second, the overall gap to the reference. At the fair, matched comparison our F1 is 72.4 percent against the reference's 79.9 percent. But that comparison is still not like-for-like: the reference trained on more than twice our data, uses an ensemble of five models where we use one, and has that anatomy cleanup stage we lack. Our number is a floor, not a verdict on the architecture.

Third, time. Sharper resolution tripled the cost of an experiment: run 2 took 12.5 hours, run 3 took 40. On a single desktop computer, that limits how fast we can iterate - which is why the plan moves training to the hospital's four-GPU machine.

10. What happens next

- Fix the ruler permanently: restate the size filter in cubic millimetres and re-score runs 2 and 3 on the identical physical standard.
- Teach the model anatomy: use an organ-mapping tool to automatically discount natural glow in the bladder, kidneys and salivary glands - aimed squarely at the false alarms on healthy scans.
- One clean experiment on the penalty balance: rerun the sharper resolution with run 2's old penalty settings, so we can finally say which change caused what.
- Add deep supervision - a training aid the reference uses that we have not adopted yet.

- Scale up: move to the hospital's four-GPU machine, train on the full thousand-scan dataset with proper cross-validation, and make the comparison to the reference truly like-for-like.

11. The bottom line

Three runs in, every headline number has improved run over run, training is completely stable, one metric has passed the published reference, and the evaluation itself has been made fairer for every run to come. The open problems are specific and understood: false alarms from natural glow, and a precision gap that the anatomy cleanup step is designed to close. The project moves to bigger hardware from a position of strength.